MAHESH REDDY MULA

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EDUCATION

National Institute of Technology, Raipur **CGPA: 8.65 / 10.0**
Bachelor of Technology in Chemical Engineering Expected May 2027

Indian Institute of Technology, Madras **CGPA: 7.50 / 10.0**
Bachelor of Science in Data Science & Applications Expected Mar 2028

TECHNICAL SKILLS

Languages & Core: Python, C++, SQL, JavaScript, HTML

Agentic AI & LLM Systems: LangGraph, LangChain, RAG, Model Context Protocol (MCP), HITL Controls

Machine Learning & Data Science: PyTorch, Scikit-learn, XGBoost, Optuna, Transformers, Deep Learning, NLP, Computer Vision

Tools, Databases & Infrastructure: PostgreSQL, FastAPI, React, Docker, GCP, Git, PyTest, Vector DBs (ChromaDB)

EXPERIENCE

Research Intern May 2025 – Jul 2025
Indian Institute of Technology Madras, Research Dept. Chennai, India

– Developed automated Python data pipelines to clean, preprocess, and analyze multi-variable experimental sensor datasets, modeling non-linear fluid dynamics.
– Engineered quantitative image analysis and statistical modeling workflows on high-resolution microscopic imaging data to measure structural distributions and validate experimental variance.

PROJECTS

Amazon ML Challenge: Multimodal Entity Extraction & Pricing | *ResNet, BERT, Boosting* Oct 2025

– Built an end-to-end data pipeline to clean, validate, and predict structured product attributes and prices from 75,000+ noisy e-commerce records.
– Engineered feature extractors using BERT Transformer embeddings and regex unit normalizers (weight, volume, dimensions) to standardize catalog metadata, fusing them with image embeddings into an aligned feature set.
– Trained gradient-boosted ensembles (XGBoost) with stratified k-fold cross-validation and Optuna tuning, achieving SMAPE 49.23 and ranking **819th out of 82,500+ teams (Top 1%)** under strict compute constraints.

OwnChatGpt: Full-Stack Agentic System & Execution Engine | *FastAPI, React, PostgreSQL, LLMs* Oct 2026

– Engineered a full-stack AI assistant with a React interface streaming real-time responses over Server-Sent Events (SSE) and a Python FastAPI backend with an LLM agent automating multi-step tool use.
– Implemented semantic document retrieval (RAG) using ChromaDB vector embeddings with source citations, and designed PostgreSQL schemas with indexing and migrations for secure session persistence.
– Integrated Model Context Protocol (MCP) servers and sandboxed shell/code tools, enforcing server-side Human-in-the-Loop approval for sensitive actions, validated with PyTest unit tests.

Agentic Blog Studio: Multi-Agent Orchestration System | *LangGraph, Python, AI Agents* Aug 2026

– Architected a multi-agent workflow using LangGraph in Python, running asynchronous fan-out sub-agents (Planner, Writer, Fact-Checker) for concurrent text generation.
– Implemented deterministic 'critic-and-revise' feedback loops with semantic assertion checks against source references, automatically identifying and rewriting hallucinations.
– Integrated LangSmith telemetry tracing to benchmark execution latency, monitor token usage, and inspect state transitions across agent graphs.

COMPETITIONS & INDUSTRY SELECTIONS

– **Tata Elxsi TellPORT (Season 3):** Earned **Pre-Placement Offer (PPO)** in 3rd year B.Tech out of 47,000+ teams for edge telematics and predictive maintenance systems design. Feb 2026
– **Nestlé–Novartis NEST 2.0:** Selected as **National Semifinalist** out of 33,300+ teams for architecting an automated clinical trial data quality and site compliance platform. Jan 2025
– **Walmart Sparkplug Summer Internship 2026:** Cleared competitive technical selection rounds and was **Shortlisted for Technical Interview**. 2026
– **Razorpay AI Buildathon 2026:** Engineered *MuleHunter*, a graph-based transaction flow tracer identifying multi-hop mule account topologies with BFS scoring. 2026
– **Amazon ML Challenge 2025:** Placed in **Top 1% (Ranked 819th / 82,500+ Teams)** for multimodal price prediction combining deep visual and NLP embeddings. Oct 2025
– **Varroc Eureka Challenge 3.0:** Earned **Pre-Placement Interview (PPI)** for automotive component technology innovation. 2025
– **Bloom Energy Be Hackfest:** Earned **Pre-Placement Interview (PPI)** in clean energy challenge tracks for applied engineering systems design. 2025
– **The Economic Times ET Gen AI Hackathon:** Earned **Pre-Placement Interview (PPI)** for generative AI application architecture. 2025
– **Stanford Online & DeepLearning.AI:** Certified in Advanced Learning Algorithms, instructed by Andrew Ng. Feb 2025

POSITIONS OF RESPONSIBILITY

Web Operations Head | *Bandipur House, IIT Madras BS* Jun 2024 – Feb 2025

– Maintained digital operations, writing SQL queries and managing election data tracking pipelines for 1,000+ student users.

Tech Executive | *Entrepreneurship Cell, NIT Raipur* Mar 2023 – Sep 2025

– Handled analytics tracking and database records for entrepreneurship events, generating automated attendance and engagement reports.